

callback Hell << Promises chains << Async Await
another

Scripting Languages.

FEBRUARY 2022
S S M T W T F S S
1 2 3 4 5 6 7 8 9 10 11 12 13
14 15 16 17 18 19 20 21 22 23 24 25 26 27
28

(003-362) Wk 02

JANUARY

MONDAY

03

- All scripting languages are programming languages.
- The scripting language is basically a language whose instructions are written for run-time environment.
- They do not require compilation step and are rather interpreted.
- It can run on multiple platforms.

Types: 1) Server-side scripting
2) Client side scripting

client side
S.L.

- ① script is executed at by web browser at user's computer
- ② cannot connect to databases which is on server
- ③ Cannot have access to files on server
- ④ Access the files & settings that are local at the users Computer

Server side
S.L.

- ① Executed by web server by that outputs the page which is to be sent to browser.
- ② Can connect to and access to the databases located on server.
- ③ Has access to the file system located on server
- ④ cannot access the settings that belong to web server.

04

(004-361) WK 02

JANUARY

AOD

TUESDAY

JANUARY 2020

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$$f(a+h) = f(a) + h \cdot f'(a)$$

- ⑤ User can block the client side scripting
- ⑥ JavaS., VBScript, etc.
- ⑤ User cannot block the server side scripting
- ⑥ PHP, JSP, ASP, ASP.NET, Ruby, Perl, etc.

ADVANTAGES:

- Easy learning
- Fast editing
- Interactivity
- Functionality
- Open Source

- APP:
- Web Application
 - System Administration
 - Games App.
 - Plugins and extensions for existing app.
 - Data Analytics
 - Testing

- DISAD:
- Slow Performance
 - Memory consumption
 - Data Security
 - Scalability

back)

callback Hell << Promises chains << Async
Await

placed callback one below another

JAVASCRIPT

JANUARY - 2022
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FEBRUARY - 2022
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(005-360) Wk 02

JANUARY

WEDNESDAY

05

• JS is a lightweight object-oriented programming language which is used by several websites for scripting webpages.

• developed by Mr. Brendan Eich in 1995

• Not a compiled language, but it is translated language.

• JavaScript translator (Embedded in browser) is responsible for translation

• It makes our webpage more lively and interactive.

• Initially called as JS and Livescript and later changed to JS.

FEATURES:

- All popular web browsers support
- follows syntax & structure of C. Thus, it is structured programming language.
- Weakly typed language.
- OOP language
- Light weight & interpreted language.
- Case sensitive

Peace is so hard to find, because it is under your nose

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JANUARY
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Applications:

- client-side validation
- dynamic drop-down menus
- displaying data and HTML
- displaying pop-up windows & dialog boxes
- displaying alerts.

CHARACTERISTICS:

- lightweight interpreted applications
- Designed for developing network-based applications
- Complementary to Java
- Integrated with HTML
- Open-source & cross platform
- Interactive interpages

ADVANTAGES:

- Immediate response
- No need to go to server hence, fast execution
- Improve usability
- Dev: get more control over the look and feel of web widgets
- Reusable

DISADVANTAGES:

- Not supported by all browsers
- Need of more quality assurance testing
- May need more time for dev.

callback hell & promise chains Async
Async

needed callback are helpful and their

2022

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100% 79% 76% 66%

JANUARY

FRIDAY

07

VARIABLES & CONSTANTS

• Variable is a name given to memory location, where we can store some value.

3. Primitive data types

- String : Sequence of characters
- Number : Numeric value
- Boolean : True or False
- Undefined
- Null : No value at all

2. Non-Primitive

- Object
- Array
- RegExp

• JS is dynamic type language hence, no need to specify type of variable.
• Type is dynamically decided by JS engine
• "var" is used to declare variable

```
var roll-no = 101;  
var name = "Shradha";  
var marks variable
```

Call back function

function greet ("name", callback)

callback hell

callback

In JS, variable can also be used without declaring it.

It automatically becomes global

The const keyword was also introduced in ES6 to create constants.

One const is initialized, we cannot change its value.

const x = 5;

x = 10; // Error! constant cannot be changed

JS Variable scopes

Scope refers to availability of variables and functions in certain parts of code.

- 1) Global scope (outside function)
- 2) Local scope (within function)

*** let is block scoped (only accessed in inner block)

DATA TYPES

- 1) Primitive
- 2) Non Primitive

Backward

back well <- promises change Async
need callback one below another
forming pyramid structure

• Function helps to avoid repetition of code and write modular code. A program can be divided into small & manageable modules called as function.

```
<script>  
  function Funct-name (Param-list)  
  {  
    statements;  
  }  
</script>
```

While defining a function, we can declare variables in the header statement of the function. These variables are known as parameters or formal arg.

JS Array?

• An array is a group of elements of same data type. All the elements in array have index numbers which starts from zero.

```
var arr = new Array ('kunal', 'Ishita', 'Sha-  
v', 'I');  
var std = ["kunal", "Ishita", "Shreyas"];
```

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Methods: length, concat(), indexOf(), lastIndexOf(), push(), pop(), sort(), reverse(), join(), splice(), slice(), toString().

var day = ["Sunday", "Monday", "Tuesday"];
delete days[1];

SPLICE:

It is used to insert new delete existing and replace existing elements by new elements in the array.

var letters = ["a", "b", "c", "d", "e", "f"];

alert(letters.splice(3,2)); // d,e (deleted)

var letters = ["a", "b", "c", "d"];

alert(letters.splice(1,2, "e", "f", "g"));

alert(letters);
// a,e,f,g,d

class
JavaScript is a blueprint for creating objects. It provides a structured way to define object properties and methods.

JS Objects

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- An object is nothing but an entity having its own state & behaviour (properties & methods).
- JS is an object-oriented language. Everything in JS is considered as an object.

Examples of Object:

- (i) Booleans (ii) Numbers (iii) Strings
- (iv) Nulls (v) Regular exp. (vi) Arrays

• We can create user defined objects in JS.

JS EVENTS

- (i) Window events : on load on abort
on unload on zoom
on resize on blur
on move on error

- (ii) Mouse events : on mousedown on mousemove
on mouseup on click
on mouseover on dblclick
on mouseout

You can not make food of this one who keeps cool

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JANUARY
THURSDAYM T W T F S S M T W T F S S
1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17 18 19 20 21 22 23 24 25 26 27 28 29 30 31(iii) Keyboard events: (i) onkeydown
(ii) onkeyup
onkeypress(iv) Form events: onsubmit onselect
onreset

JSON

- Java's Object notation
- Text format which is used for storing data
- Originated from JS
- Self describing
- Language independent
- Supports array, object, string, number, & null

JS Arrow Functions

Introduced in ES6 version

```
let x = function(x, y) {
  return x * y;
}
```

⇒ let x = (x, y) => x * y

Do not expect your doc attention. Give it instead

babay;

callback hell < Promises, Async/Await

No need callback and before another

*** Timely job hoga tab Asynchronous

DEC-2022

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JANUARY

14

let greet = (c) => console.log("Hello");

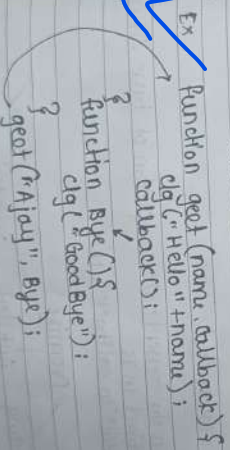
let greet = x => console.log(x);

let welcome = (age < 18) ? () => console.log("baby") : () => clg("Adult");

CALLBACK

A callback function is a function i.e. passed as an argument to another function and executed later.

- A function can accept another function as a parameter.
- callback allow one function to call another at a later time

Ex: 

```
function greet(name, callback) {  
  clg("Hello" + name);  
  callback();  
}  
function Bye() {  
  clg("Good Bye");  
}  
greet("Ajay", Bye);
```

✓ Callback Hell is a situation in JavaScript where multiple nested callbacks make the code difficult to read, debug, and maintain. It occurs when executing asynchronous operations that depend on the results of previous operations, leading to deeply nested and hard-to-follow code.

Callback Hell << promise chains << async await

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1 2 3 4 5 6 7 8
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Callback Hell

- Nested callbacks stacked below one another forming a pyramid structure

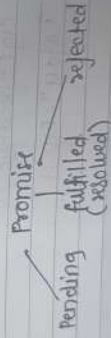
```
getData(1, () => {  
  console.log("getting data 2...");  
  getData(2, () => {  
    console.log("getting data 3...");  
    getData(3, () => {  
      console.log("getting data 4...");  
      getData(4);  
    });  
  });  
});
```

- When multiple async operations depend on each other.

PROMISES

- Promise is for "eventual" completion of task. It is an object in JS.
- It is held in callback hell.

16 Sunday



Remember! You are special. No one can play your role better than you.

[illegible]

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JANUARY
MONDAY

fulfilled

Reject

promise, then $(res) = \{ \dots \}$ promise, catch $(err) \{ \dots \}$

CODE 7

```
const getPromise = () => {  
  return new Promise((resolve, reject) => {  
    console.log("I am promise")  
    resolve("Success")  
    // reject("Error")  
  })  
}
```

```
let promise = getPromise();
promise.then((res) => {
  log("Promise Fulf.:", res);
});

promise.catch((err) => {
  log("Rejected with:", err);
});
```

★★ promise. Finally is used when either resolved successfully or rejected.

Callback Rndn
 Rndn greet ("wund", callback)
 dia callback
 asked call
 ing

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 TUESDAY

```

1 console.log("fetching 1...");
2 let p1 = asyncFunc();
3
4 p1.then(res => {
5   console.log("fetching 2...");
6   let p2 = asyncFunc2();
7   p2.then(res => {});
8 });
  
```

Promise
 Chaining

ASYNCAWAIT

• async function always return a promise

async function myFunc() { ... }

• await pauses the execution of its surrounding
 async function until promise is settled.

Async:

- Always return promise
- To represent that the function is asynchronous

Avoid:

- Await is used inside async
- Pauses the async function until the promise return a result.

The four things that have spoiled your life are: "I", "Me", "You" and "Ours". Forget them.

check examples from gpt

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WEDNESDAY

Asynchronous method to be executed in a seemingly synchronous way.

✓ Although the operation is async, it seems that the operation is executed in a synchronous way.

JS ERROR HANDLING

• It is a loosely-typed language. It does not give compile-time errors. So sometimes, you will get a runtime error for accessing an undefined variable or calling undefined function, etc.

• JS provides error-handling mechanism to catch runtime errors using try-catch-finally block, similar to other languages like Java or C#.

• try: wrap suspicious code that may throw an error in try block.

• catch: write code to do something in catch block when an error occurs.

• finally: always be executed regardless of the occurrence of an error.

There was never a right or problem that could defeat sunrise or hope.

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AJAX

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10 11 12 13 14 15 16 17 18 19 20 21 22 23 24 25 26 27 28 29 30 31

Asynchronous javascript in XML (AJAX)

- AJAX is not a programming language. Rather, it's a set of existing technologies.
- Helps in fetching data asynchronously without interfering with the existing page.
- No page reload/refresh
- Modern websites uses JSON instead of XML for data transfer. It means that web app could send & receive data from the web server without refreshing the page.

BENEFITS:

- Using it, you can create better, faster & more user-friendly web app.
- Based on JS, CSS, HTML & XML.

WHERE:

- Login forms
- Auto complete
- Rating & Voting
- Updating with user content

ImagePlaceholder is like a placeholder. It does not allow to use

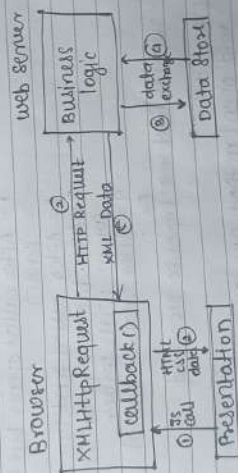
those hell << promises chains << Async Await
 Async call back are helpful and then
 Async call back are helpful and then

ADV:

- Reduces server traffic & increases speed
- It is responsive & time taken is also less
- Form validation
- Bandwidth usage can be reduced
- Async call can be made, this reduces the time for data arrival.

DISADV:

- Open source
- You can only access info. from web host that serves pages.



Callback function
 function greet (username, callback) {
 callback (username);
 }

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 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17 18 19 20 21 22 23 24 25 26 27 28 29 30 31

Purpose of XMLHttpRequest:

- Interact with servers asynchronously
- It sends data in background to server.
- It requests data from the server.
- It receives data from the server.
- Updates data without reloading page

PROPERTIES:

onreadystatechange → Event Handler based on ready state

readyState → 0 = obj created but before open
1 = After called open() but before send()

2 = After called send()
3 = After browser established a conn. with server
4 = After request complete

responseText → Response received

METHODS:

void open (method, URL, async)
 void send () → get request
 void send (string) → post request

REPLICATES	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25	26	27
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1xx → Informational
2xx → Successfull
3xx → Redirection
4xx → Client Error
5xx → Server Error

QUERY

- It is a javascript library
 - Easy to learn
 - Purpose is to make it much easier to use JS on your website.
 - "Write less, do more"
 - Fast. Feature-rich JS library
 - DOM manipulation, event handling, animation & Ajax request
 - Write less code achieve more functionality
- ADVANTAGES:
- works on all platform
 - large & Adv. set of functionality
 - lightweight
 - supports AJAX tech
 - offers event handling
 - Animation & DOM manipulation

AT

- works on all platforms
- Java & Adv. Set of Functionality
- Lightweight
- Supports AJAX Tech
- Offers event handling
- Animath & DOM manipulation

Callback function
 function greet ("world", callback) {
 // callback
 }
 greet ("world", callback);

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TUESDAY
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DISADV.

- Must be installed from internet to use
- coding must be in support of JavaScript, AJAX, ASP, PHP

WAYS TO USE :

- Download
- CDN (Google OR MS)

SYNTAX :

$\$(selector).action$

- $\$$ → sign to define/access jQuery.
- (selector) → "query" HTML element
- action() → action() to be performed

- All jQuery methods are inside document ready and to wait for the doc. to be fully loaded and ready before working with it.

$\$(document).ready(function() {$
 // jQuery methods
 $\})$;

The beauty of all things lies in your ability to appreciate them.

about Well << Promising drinks << Aspic
 need call back one below another
 common in mind structure

ex. \$(document).ready(function() {

\$("p").css("background-color",
 "pink");

});

\$(this).hide() - hides current element
 \$("p").hide() - hides all <p> elements
 \$(".test").hide() - class = "test"
 \$("#test").hide() - id = "test"

ANIMATION:

- show()
- hide()
- fadeOut()
- fadeIn()
- slideUp()
- slideDown()

• append(" Nagpur ");
 • remove()
 • empty()
 • css

DOM Manip.
 using JQUERY